

DAY_11

❖ Wireless Standard & Configuration : Wireless Setup, Channel Selection & Avoiding Interference, Wireless Security, MAC Filtering & Access Control Lists

1. Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

ANS :

- To understand basic wireless configuration, channel selection, wireless security, MAC filtering, and router ACLs. The practical covers setting up a Wi-Fi network, reducing interference, securing the wireless network, controlling device access, and restricting access to a specific destination.

▪ Network Setup

Setting	Value
Router	Home Wi-Fi Router
SSID	College_WiFi
Wireless Band	2.4 GHz
Default Channel	Auto
Security	WPA2-Personal
Password	College@WiFi2026

▪ Procedure

1. Power on the Wi-Fi router.
2. Connect your laptop or phone to the router.
3. Open the router's administration page.
4. Go to **Wireless/Wi-Fi Settings**.
5. Note the SSID and current channel.

2. Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks. Hint: Look for overlapping channels and pick one with the least overlap.

ANS :

➤ **Use Wi-Fi Analyzer**

- Install a Wi-Fi analyzer application on your phone and scan nearby networks.

Channel	Nearby Networks	Interference
1	6	Medium
6	2	Low
11	7	High

➤ **Procedure**

1. Open the Wi-Fi analyzer.
2. Scan nearby wireless networks.
3. Check which channels are heavily occupied.
4. Select the channel with the least interference.
5. Configure that channel manually on the router.
6. Reconnect the wireless device.
7. Perform the scan again.

3. Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

ANS :

➤ **Recommended configuration:**

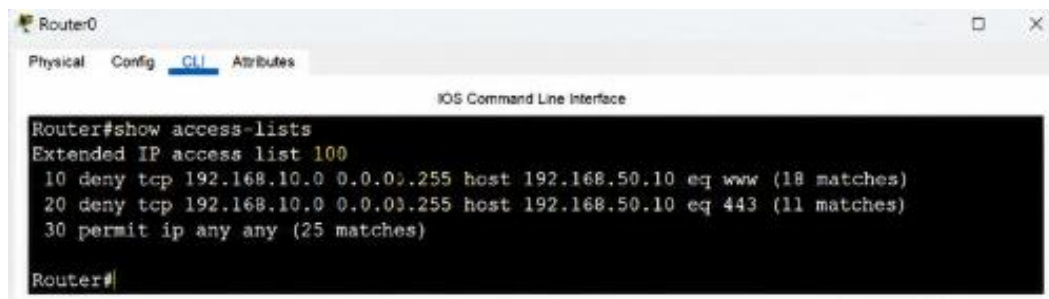
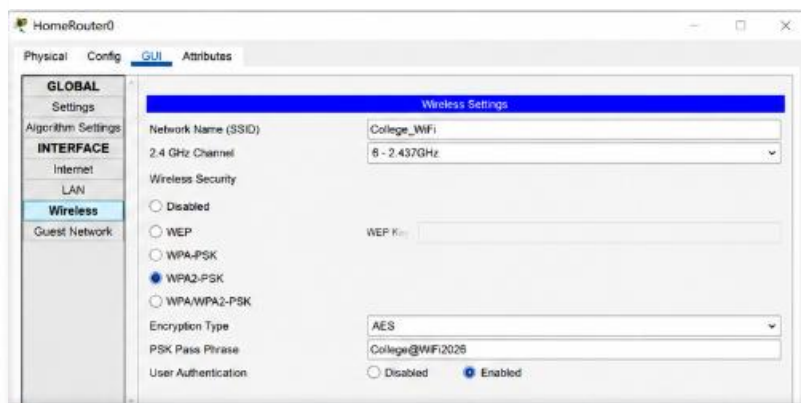
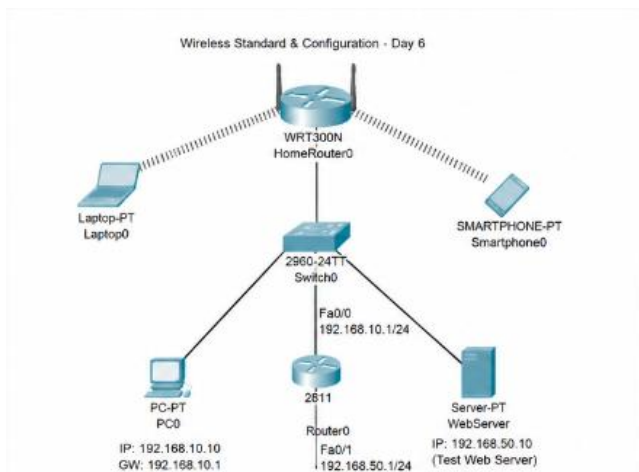
- Security Mode: WPA2-Personal
- Encryption: AES
- SSID: College_WiFi
- Password: College@WiFi2026

- If your router supports WPA3:
 - Security Mode: WPA3-Personal
 - Use a strong password containing uppercase/lowercase letters, numbers, and symbols.
- Example:
 - College@WiFi#2026!

▪ Test the Security

On the laptop or phone:

1. Forget the existing Wi-Fi network.
2. Search for College_WiFi.
3. Select the network.
4. Enter the new password.
5. Connect.



4. Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result. Hint: You can find your device's MAC address in the network settings.

ANS :

- MAC filtering allows the router to permit or deny devices based on their **MAC address**.

➤ **Find the Device MAC Address**

- On Windows:
ipconfig /all
- Look for:
Physical Address
- Example:
A4-B1-C1-22-33-44

➤ **Test with an Unauthorized Device**

Device	MAC Address	Status
Laptop	A4:B1:C1:22:33:44	Allowed
Phone	78:25:AD:55:66:77	Denied

5. Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.

ANS :

- An ACL can be used to control traffic based on IP addresses, protocols, ports, and other packet characteristics.
- For a simple **Cisco Packet Tracer lab**, it is better to block a specific **destination IP address** representing the website/server.

➤ Important Practical Note

- A normal router ACL does **not reliably block a website by its name**, such as www.youtube.com, because modern websites can use multiple changing IP addresses and HTTPS. For a classroom demonstration, use a known test-server IP or a controlled web server.

➤ Example Extended ACL

- Enter:
Router> enable
Router# configure terminal
- Create the ACL:
Router(config)# access-list 100 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq 80
Router(config)# access-list 100 deny tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq 443
Router(config)# access-list 100 permit ip any any
- Apply it to the interface:
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip access-group 100 in
Router(config-if)# exit

➤ Practical Observation Table

Task	Configuration	Observation
Wireless Setup	SSID = College WiFi	Device connected successfully
Channel Selection	Channel changed from Auto to 6	Lower interference observed
Wireless Security	WPA2/WPA3 enabled	Correct password required
MAC Filtering	Authorized MAC added	Authorized device connected
ACL	Test server blocked	Web access was denied